JoongWon Shin

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Education

Seoul National University Mar. 2022 – Present
Undergraduate Student in Computer Science and Engineering
Completed Credits: 91
Overall GPA: 4.03 / 4.30 (3.88 / 4.00) | Major GPA: 4.03 / 4.30
Mandatory Military Service: May 2023 – Oct. 2024 (completed)

Honors and Scholarships

National Scholarship for Science and Engineering Awarded 2022
Selected for a four-year full-tuition scholarship from the Korea Student Aid Foundation based on Korea's College Scholastic Ability Test (CSAT) scores

Interests

- Hardware-Software Co-Design for AI Systems
- AI Accelerators
- Efficient LLM Inference

Research Experience

Undergraduate Research Intern, ARC Lab, Seoul National University Jun. – Sep. 2026
Advised by Prof. Jae W. Lee

- Contributed to research on CXL-PNM-based hardware/software co-design for efficient LLM serving.
- Contributed to research on efficient inference for SSM and hybrid-attention language models.

Publications

SketchSSM: Write to the Full State, Read from a Compact Sketch
Omin Kwon, **JoongWon Shin**, Minseo Kim, Kurt Keutzer, Sehoon Kim*, Jae W. Lee*
NeurIPS 2026 LCFM Workshop **(Oral)**
Extended version under review.

Projects

Wastory, WaffleStudio Software Project 2025

- Collaborated on a team-based application development project using Git-based workflows and code review.

Selected Coursework

Systems and Architecture:	Computer Architecture, Logic Design, System Programming, Hardware Systems Design *In progress:* Multicore Computing, Operating Systems, Mobile Computing and Its Applications
AI and Machine Learning:	Computer Vision *In progress:* Computer Convergence Application, Basics of Deep Learning
Algorithms and Theory:	Data Structures, Algorithms, Discrete Mathematics, Automata Theory
Mathematics and Statistics:	Engineering Mathematics I–II, Statistics
Programming and Software:	Computer Programming, Programming Practice, Software Development Principles and Practice

Technical Skills

Programming:	C, C++, Python, Java, Swift
Hardware Description:	Verilog, Amaranth
Tools and Frameworks:	Git, PyTorch